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Reservoir Fluid Compositions – A Result of Chemical Reaction Equilibrium and Gravity Segregation

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Abstract

Petroleum reservoir fluids exhibit characteristic compositional patterns. The paper analyses whether these patterns are due to the fluids being in a state of chemical reaction equilibrium in the reservoirs. It is found that reservoir fluids can be divided into three groups, those at chemical reaction equilibrium, those not at equilibrium because they are under strong influence of a vertical compositional gradient, and finally those that have not yet reached chemical reaction equilibrium. The theory and model parameters for calculation of fluid compositions at chemical reaction equilibrium are presented, and examples are given of reservoir fluids in equilibrium as well as reservoir fluids that are not in equilibrium. An analysis is performed of reservoir fluids from a North Sea field, which field is shown to contain fluids in chemical reaction equilibrium, fluids that are not in equilibrium because they are affected by a vertical compositional gradient, and a fluid that has not yet reached chemical reaction equilibrium. Fluids of the latter type have a lower concentration of C_1 than the equilibrium compositions and correspondingly higher concentrations of C_2 - C_5 . The theory of chemical reaction equilibrium allows, for both equilibrium and non-equilibrium fluids, to estimate an approximate composition of a reservoir fluid solely from the reservoir pressure and temperature, and the density of the fluid under these conditions.

Introduction

Reservoir fluid compositions exhibit certain patterns, one being an approximate linear relationship between the logarithm of the mole percents of the C_{7+} carbon number fractions and the carbon numbers [1,2]. As illustrated in Figure 1, petroleum reservoir fluids are formed by decomposition of the remains of plants and animals, primarily marine organisms. It has been proposed that the patterns seen in oil and gas compositions are due to the fact that the breakdown into smaller molecules occurred in such a way that the compositions formed were the chemically optimal ones under the given conditions and with the available amounts of hydrogen and carbon atoms [3,4]. That would mean a reservoir fluid in a state of chemical reaction equilibrium. This theory has proven successful for shale reservoir fluids [5]. These are found in tight formations leaving the molecules with limited ability to move. In most other reservoirs, molecules can move more freely, and gravity and a vertical temperature gradient will make the concentration of heavy

components increase with depth. A field with a compositional gradient is neither in phase equilibrium nor in chemical reaction equilibrium, but the deviation from equilibrium is not necessarily large if the reservoir is of limited depth.

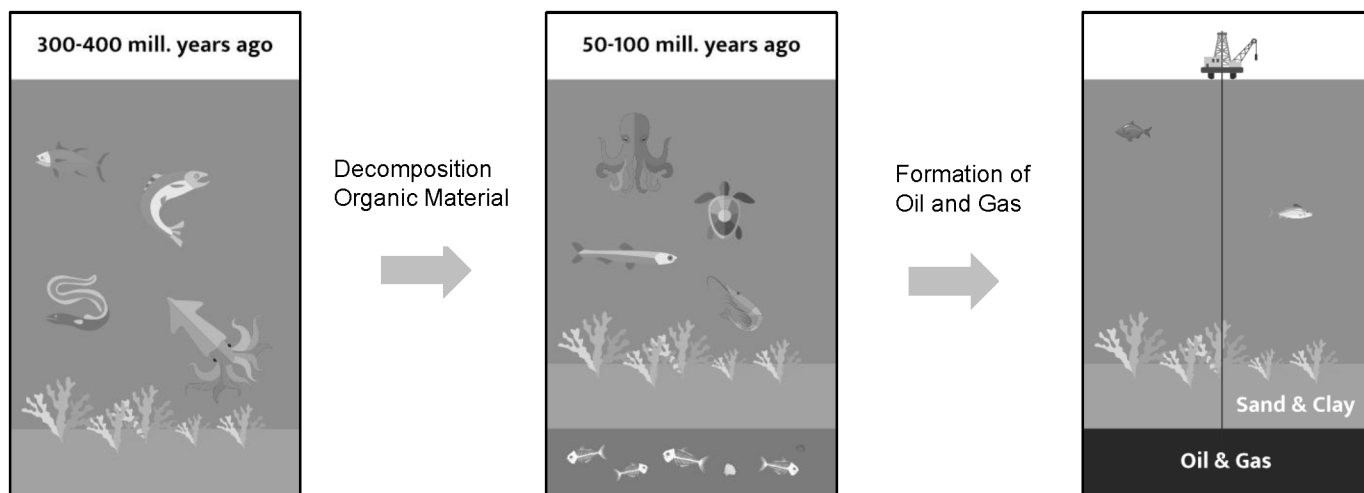


Figure 1—Illustration of how oil and gas were formed.

Also, for fields with a significant compositional gradient, there will be information to be gained by comparing a measured reservoir fluid composition with the fluid composition at chemical reaction equilibrium. A fluid sample from the upper part of a reservoir will have a lower concentration of C_{7+} components compared to the composition at chemical reaction equilibrium, while the opposite will be the case for a sample from the lower part of the reservoir. Similar fluid compositions are often seen in neighboring fields, even though the assessment is that the fields do not communicate but are in separate compartments. If it can be shown for such reservoirs that the fluids are at chemical reaction equilibrium, it would explain why the fluids are almost identical, and this might help rule out possible communication.

The organic material that led to the formation of oil and gas started with large molecules that gradually broke down into smaller molecules. Methane (C_1) is the hydrocarbon with the lowest molecular weight and chemically represents a favored end product. It is therefore to be expected that a reservoir fluid that is not in chemical reaction equilibrium will have a lower concentration of C_1 than the equilibrium composition, and that the deviation between the C_1 concentrations in an equilibrium composition and an actual non-equilibrium composition will provide a measure of how far the actual composition is from equilibrium. Some reservoir fluid compositions are affected by biodegradation, which is caused by bacteria, algae or fungi that have consumed lighter normal paraffins. The paper does not cover this type of fluid.

Chemical Reaction Equilibrium Theory

As sketched in Figure 2, an assumption of chemical reaction equilibrium means that a petroleum reservoir is considered to be a kind of chemical reactor, where some hydrocarbon components are broken down and others are formed to achieve the oil or gas composition that is optimal considering pressure (P), temperature (T), and the ratio of hydrogen (H) atoms to carbon (C) atoms.

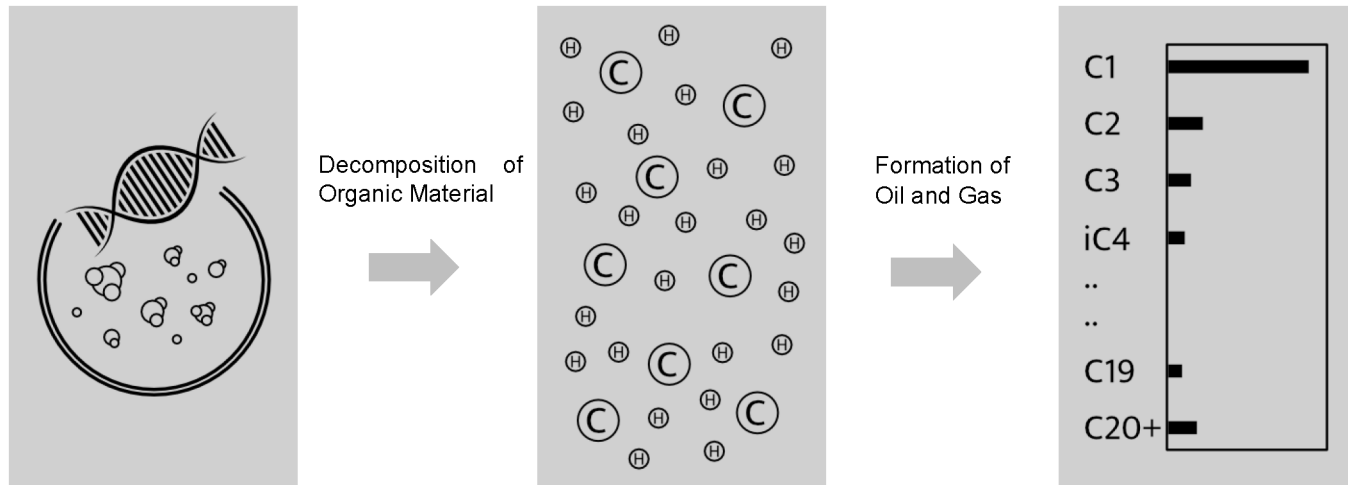


Figure 2—Illustration of the formation of oil and gas at a molecular/atomic level. Organic material from the remains of plants and animals was broken down and formed the hydrocarbon molecules that make up oil and gas.

A closed system at a constant temperature, T , and pressure, P , will try to minimize its total Gibbs free energy, G . The principle is known from phase equilibrium calculations, where the phase quantities and phase compositions for the considered total composition adjust themselves to minimize G . If chemical reactions can take place, the molar composition is an additional variable, and a composition will form that at the given pressure and temperature minimizes G . Disregarding inorganic components, the equilibrium composition of a petroleum reservoir fluid will be that of the lowest G for the given H/C ratio. The total Gibbs free energy, G^t , of a single-phase system at temperature, T , and pressure, P , containing N different species, can be expressed as follows:

$$(G^t)_{T,P} = G(n_i) \quad i = 1, 2, \dots, N \quad (1)$$

where n_i is the number of moles of component i . The task is to find the set of n_i 's that minimizes G^t at T and P subject to the constraints of the material balance. The first step is to formulate the material balance. The concentrations of the individual hydrocarbon components may vary because of chemical reactions, but the total number of atoms of each element will remain constant. In the following, the sub-index k is used to identify atomic species. A_k is the total number of atoms of type k and a_{ik} stands for the number of atoms of type k in molecule i . The material balance for each element, k , may then be written

$$\sum_{i=1}^N n_i a_{ik} - A_k = 0 \quad k = 1, 2, \dots, K \quad (2)$$

where K is the total number of elements contained in the fluid. For a system consisting solely of hydrocarbons, K will be equal to 2 and the elements are carbon (C) and hydrogen (H).

To facilitate the determination of the number of molecules of each chemical species, Lagrange multipliers, λ_k , one for each element, are introduced by multiplying each element balance in Equation (2) by its Lagrange multiplier and summing over all k 's:

$$\sum_{k=1}^K \lambda_k \left(\sum_{i=1}^N n_i a_{ik} - A_k \right) = 0 \quad (3)$$

A new function, F , is formed by adding the sum in Equation (3) to G^t in Equation (1):

$$F = G^t + \sum_{k=1}^K \lambda_k \left(\sum_{i=1}^N n_i a_{ik} - A_k \right) \quad (4)$$

The new function, F , will have the same value as G^t , because the summation term is zero, but the partial derivatives of F and G^t differ because F incorporates the material balances. The minimum value of both F and G^t may be found by differentiation of F with respect to n_i . At the minimum, this partial derivative is zero:

$$\left(\frac{\partial F}{\partial n_i}\right)_{T,P,n_j} = \left(\frac{\partial G^t}{\partial n_i}\right)_{T,P,n_j} + \sum_{k=1}^K \lambda_k a_{ik} = 0 \quad (5)$$

The derivative on the right-hand side is the definition of the chemical potential of component i , μ_i , which enables derivation of the following set of equations:

$$\mu_i + \sum_{k=1}^K \lambda_k a_{ik} = 0 \quad i = 1, 2, \dots, N \quad (6)$$

The chemical potential of component i may also be written:

$$\mu_i = \Delta G_{fi}^0 + RT \ln(f_i) = \Delta G_{fi}^0 + RT \ln(z_i \varphi_i P) \quad (7)$$

where ΔG_{fi}^0 is the standard Gibbs energy of formation of component i at the temperature, T , R is the Gas Constant, and f is fugacity. In the last term, the fugacity of component i is rewritten to a product of its mole fraction, z_i , its fugacity coefficient, φ_i , and the pressure, P . The following set of equations is reached by combining Equations (6) and (7):

$$\Delta G_{fi}^0 + RT \ln(z_i \varphi_i P) + \sum_{k=1}^K \lambda_k a_{ik} = 0 \quad i = 1, 2, \dots, N \quad (8)$$

The hydrocarbons from C_1 to C_6 are either n -paraffins or iso-paraffins for which the number of hydrogen and carbon atoms for carbon number i can be calculated from

$$(H\text{-atoms})_i = 2 \times Cn_i + 2 \quad i = 1, 2, \dots, 6 \quad (9)$$

$$(C\text{-atoms})_i = i \quad i = 1, 2, \dots, 6 \quad (10)$$

Using these two relations, Equation (8) may for the hydrocarbons from C_1 to C_6 be rewritten to

$$\Delta G_{fi}^0 + RT \ln(z_i \varphi_i P) + \lambda_C \times i + \lambda_H \times (2 \times i + 2) = 0 \quad i = 1, 2, \dots, 6 \quad (11)$$

which may be rearranged to

$$RT \ln(z_i) = -\Delta G_{fi}^0 - RT \ln(\varphi_i P) - 2\lambda_H - i \times (\lambda_C + 2\lambda_H) = 0 \quad i = 1, 2, \dots, 6 \quad (12)$$

The molecular weight, M_i , of C_{7+} carbon number fraction i may be approximated as [1]:

$$M_i = 14 \times i - 4 \quad i = 7, 8, \dots, N \quad (13)$$

where N is the highest carbon number fraction considered, which in this work will be 80. The atomic weight of carbon is 12 and that of hydrogen is 1. By rearranging Equation (13) to

$$M_i = 12 \times i + (2 \times i - 4) \quad i = 7, 8, \dots, N \quad (14)$$

it is seen that the dependence of molecular weight on carbon number in Equation (13) corresponds to $(2 \times i - 4)$ hydrogen atoms per carbon number, which leads to the following relation for the C_{7+} components:

$$RT \ln(z_i) = -\Delta G_{fi}^0 - RT \ln(\varphi_i P) + 4\lambda_H - i \times (\lambda_C + 2\lambda_H) = 0 \quad i = 7, 8, \dots, N \quad (15)$$

Depth Gradient Theory

At thermodynamic equilibrium in a closed system with negligible height differences and no chemical reactions occurring, each component will have the same fugacity (f) throughout the system. In a petroleum reservoir, significant height differences may exist, and for an isothermal reservoir, the equilibrium criterion must be modified to the following relation between the fugacity of component i at depths h and h^0 [7]

$$T \ln(f_i^h) - T \ln(f_i^{h^0}) = \frac{M_i}{R} g(h - h^0) \quad i = 1, 2, \dots, N \quad (16)$$

N is the number of components, M molecular weight, g gravitational acceleration, T temperature, and R the gas constant. The fugacity of component i is found as follows:

$$f_i = \phi_i z_i P \quad i = 1, 2, \dots, N \quad (17)$$

where z_i is the mole fraction of component i, P is pressure, and ϕ_i the fugacity coefficient of component i, which quantity may be derived from an equation of state. Equations (16) and (17) may be solved for the molar composition (z_i^h , $i = 1, 2, \dots, N$) in depth h if the pressure p^h and the composition ($z_i^{h^0}$, $i = 1, 2, \dots, N$) in the reference depth, h^0 , are known.

A petroleum reservoir with a positive vertical temperature gradient will not be at thermodynamic equilibrium. There will be a circulation of molecules from the warmer bottom to the colder top and back. If all heat transport is tied to molecular transport and the system is stationary, meaning an unchanged concentration profile with time, the relation between the fugacities at depths h and h^0 can according to Haase [8,9] be expressed as follows

$$(T + dT) \ln(\phi_i^h z_i^h P^h) - T \ln(\phi_i^{h^0} z_i^{h^0} P^{h^0}) = \frac{M_i}{R} g(h - h^0) - \frac{M_i}{R} \left(\frac{H^{abs}}{M} - \frac{H_i^{abs}}{M_i} \right) \frac{dT}{T} \quad i = 1, 2, \dots, N \quad (18)$$

H^{abs} is the molar absolute enthalpy of the mixture. dT in Equation (18) is the (differential) temperature difference between heights h and h^0 . $\tilde{H}_i^{abs}$ is the absolute partial molar enthalpy of component i. The absolute enthalpy of a component is calculated as the sum of the ideal gas enthalpy and the residual enthalpy

$$H^{abs} = H^{ig} + H^{res} \quad (19)$$

The residual term can be expressed as

$$H^{res} = -RT^2 \sum_{i=1}^N z_i \frac{\partial(\ln \phi_i)}{\partial T} \quad (20)$$

and the derivative of the fugacity coefficient can be found from the equation of state. The below expression is used for the ideal gas enthalpy at the temperature T

$$H^{ig}(T) = \sum_{i=1}^N z_i H_i^{ig}(T_{ref}) + \sum_{i=1}^N z_i \int_{T_{ref}}^T C p_i^{ig} dT \quad (21)$$

where $H_i^{ig}(T_{ref})$ is the absolute ideal gas enthalpy of component i at the temperature, T_{ref} , which temperature may be chosen to 0°C. $C p_i^{ig}$ in Equation (21) may be calculated from a 3rd degree polynomial in absolute temperature

$$C p_i^{ig} = C_{1,i} + C_{2,i} T + C_{3,i} T^2 + C_{4,i} T^3 \quad (22)$$

Poling et al. [10] have tabulated the coefficients C_1 - C_4 for defined components. For C_{7+} components, C_1 - C_4 may be calculated as outlined by Kesler and Lee [11].

Gibbs Energies of Formations

Table 1 shows the Gibbs energies of formation used for hydrocarbon components C_1 - C_6 [6]. For C_{7+} components, the following correlation is used with the coefficients a_1 - a_4 given in Table 2.

$$\Delta G_{fi}^0 = a_1 + a_2 \times Cn + (a_3 + a_4 \times Cn) \times (T - 298.15) \quad (23)$$

With these coefficients, it is taken into consideration that the C_{7+} carbon number fractions are mixtures of paraffinic, naphthenic and aromatic components.

Table 1—Gibbs energies of formation for C₁–C₆ hydrocarbon components [6].

Component	ΔG_f^0 at 298.15 K Joule/mol	$\frac{\partial(\Delta G_f^0)}{\partial T}$ Joule/(mol K)
C ₁	-50,460	108.0
C ₂	-32,608	209.1
C ₃	-21,999	322.8
iC ₄	-7,897.7	431.7
nC ₄	-10,782	440.3
iC ₅	1944.3	537.1
nC ₅	726.38	545.7
C ₆	10,676	633.1

Table 2—Coefficients in Equation (23) for Gibbs energies of formation for C₇₊ components in Joule/mol.

Coefficient	Value of coefficient
a ₁	81,805
a ₂	10,789
a ₃	-155.75
a ₄	91.537

Simulation of Equilibrium Compositions

To simulate a composition in chemical reaction equilibrium, [Equations \(12\)](#) and [\(15\)](#) must be solved. For a known molar composition (z_i , $i=1,2,\dots,N$), the fugacity coefficients can be calculated from an equation of state provided equation of state parameters (T_c , P_c , acentric factor, volume correction, and binary interaction parameters) are known for each component. For a given H/C ratio, the only unknown variables are the Lagrange multipliers, λ_C and λ_H .

The following step-by-step calculation procedure was used to test whether a fluid sample of known composition reported to C₂₀₊ was in chemical reaction equilibrium at reservoir conditions:

1. Characterize the fluid composition [2] for the applied equation of state. Do not lump C₇–C₁₉. Lump C₂₀₊ to one pseudo-component.
2. Assume initial values for λ_C and λ_H .
3. Calculate the fugacity coefficients, ϕ_i , of each component using the equation of state.
4. Determine a new fluid composition to C₁₉ by solving [Equations \(12\)](#) and [\(15\)](#).
5. Make a linear best fit to ln(mol%) versus carbon number for the C₇–C₁₉ components in the simulated composition. Simulate the C₂₀–C₈₀ mol%'s by extrapolation of this line.
6. After the first round, return to 3.
7. In subsequent rounds, check whether the estimated fluid composition is unchanged from the previous round, whether the component mole fractions sum to 1, and whether the fluid has the correct H/C ratio.
8. If not, adjust λ_C and λ_H and return to 3.
9. The equilibrium composition is found when the criteria in 7. are met.
10. Compare the composition of the equilibrium fluid with the composition of the reservoir fluid that is being investigated for possible chemical reaction equilibrium.

At the time the first sample is taken from a new reservoir, the fluid (in-situ) density and the reservoir pressure and temperature will often be the only information available, and the H/C ratio is unlikely to be known. Based on in-situ fluid densities for 32 reservoir fluid compositions with H/C ratios ranging from 1.91 to 3.00, a correlation was developed to predict the H/C ratio from in-situ fluid density, ρ , pressure, P , and temperature, T . The correlation is found to be reliable for fluid compositions with up to 5 mol% of non-hydrocarbons and takes the form

$$H/C \text{ Ratio} = A_1 + A_2 \times \rho + A_3 \times T \times \rho + \frac{A_4 \times P}{\rho} \quad (24)$$

Table 3 shows the values of the constants $A_1 - A_4$ for T in K, P in bar, and ρ in g/cm^3 .

Table 3—Coefficients in Equation (24) for T in K, P in bar, and ρ in g/cm^3 .

Coefficient	H/C ratio
A_1	3.4145
A_2	0.29850
A_3	-6.6598×10^{-3}
A_4	2.1536×10^{-4}

The calculation procedure described above requires knowledge of the concentrations of non-hydrocarbons (N_2 , CO_2 and H_2S), which are considered inert components not taking part in chemical reactions. In many cases, approximate concentrations of these components will be known from nearby fields.

Equilibrium Compositions

Table A.1 in Appendix A shows four examples of reservoir fluids whose compositions agree well with the simulated equilibrium compositions and for which the assessment is that they are in chemical reaction equilibrium at reservoir conditions. The good agreement is illustrated in Figure 3, which shows bar diagrams comparing measured compositions with equilibrium compositions. The fugacity coefficients were calculated using the volume corrected SRK equation [12,13].

Had the reservoir fluid composition been unknown and the task been to estimate the equilibrium composition with reservoir temperature, reservoir pressure, and reservoir fluid density as the only input, the procedure would be:

1. Estimate the H/C ratio from Equation (24) using the coefficients in Table 3.
2. Assume a known composition with approximately the same H/C ratio and characterize it for the chosen equation of state.
3. Estimate the composition that satisfies chemical reaction equilibrium.
4. Perform new characterization, this time of the simulated equilibrium composition.
5. Estimate the composition that satisfies chemical reaction equilibrium with the fluid from 4. as the starting point.

As can be seen from Table A.1 in Appendix A, there are some deviations between the measured and simulated mol%'s of the individual C_{7+} fractions. This is to be expected, as these may have varying contents of paraffinic, naphthenic and aromatic components, which the coefficients in Equation (23) for Gibbs energy of formation for C_{7+} fractions (Table 3) do not take that account. However, a good agreement is seen for the total C_{7+} fraction (gray-scaled in Table A.1). In case of larger deviations between the simulated and the measured total C_{7+} mol%, this may be due to a non-representative fluid sample, such as a sample taken in the two-phase region.

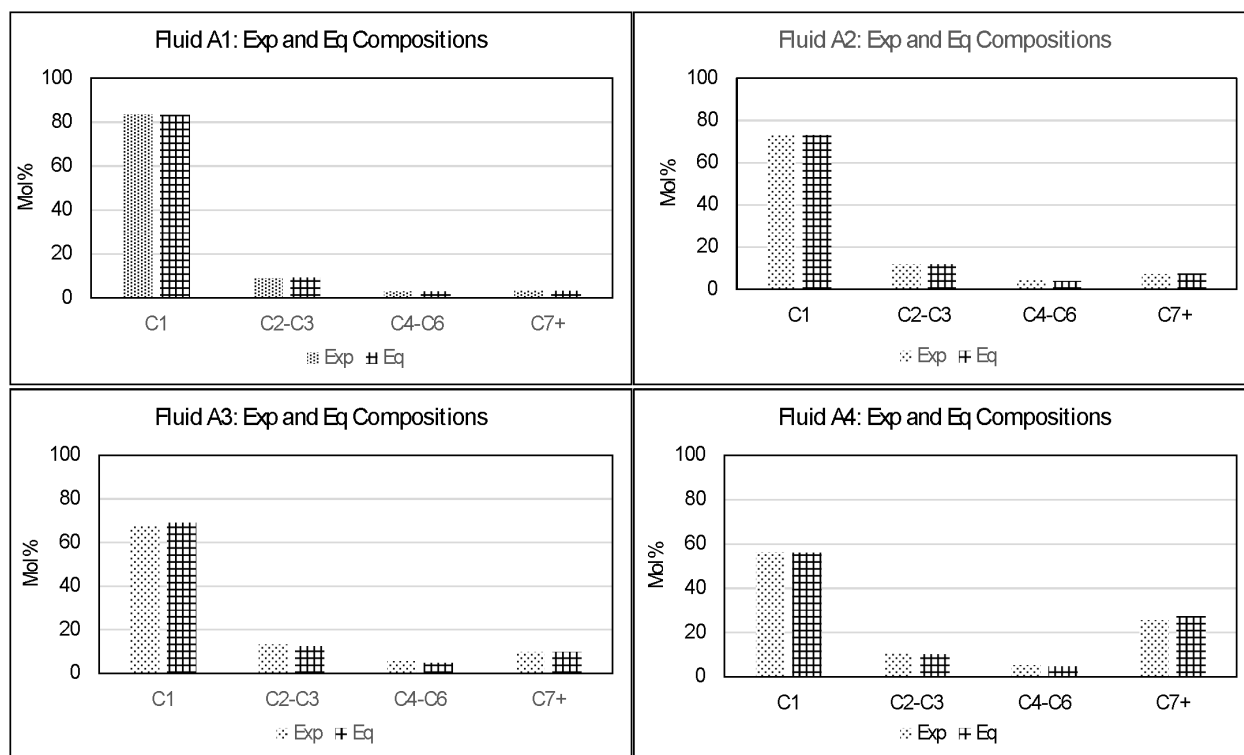


Figure 3—Bar diagrams showing measured (Exp) compositions and simulated equilibrium (Eq) compositions for four reservoir fluids that are considered being at chemical reaction equilibrium at reservoir conditions. Detailed compositions can be found in [Table A.1](#) in [Appendix A](#).

Reservoir Fluids Not at Chemical Reaction Equilibrium

Not all reservoir fluids are at chemical reaction equilibrium, but this does not mean that their compositions are random. The compositions may deviate from equilibrium because of a vertical compositional gradient, or the fluids may not yet have reached chemical reaction equilibrium, which can take millions of years. The change in composition will hardly be measurable within the lifetime of a field. [Table B.1](#) in [Appendix B](#) shows two examples of fluids that are not at chemical reaction equilibrium. [Figure 4](#) shows bar diagrams with measured and equilibrium compositions for the same two fluids. In the simulations, the fugacity coefficients were calculated using the volume corrected SRK equation [12,13]. The C_1 concentrations in the reservoir fluids are lower than in the equilibrium fluids, while the C_2 - C_6 concentrations are higher. This suggests that the fluids are in the final stage towards chemical reaction equilibrium, where primarily conversion of C_2 - C_6 to smaller molecules, primarily C_1 , takes place.

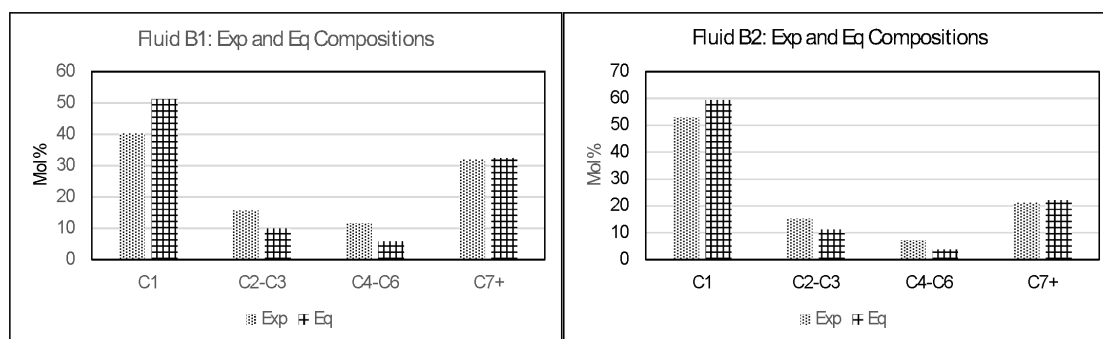


Figure 4—Bar diagrams showing measured (Exp) compositions and simulated equilibrium (Eq) compositions for two non-equilibrium fluids for which detailed compositions can be found in [Table B.1](#) in [Appendix B](#).

The final step in the conversion process, which ultimately yields an equilibrium composition, is illustrated in Figure 5. On the left, some C₂-C₅ molecules are highlighted to show that these components are present in too large numbers. To reach equilibrium, some of them will be broken down into smaller molecules as illustrated on the right.

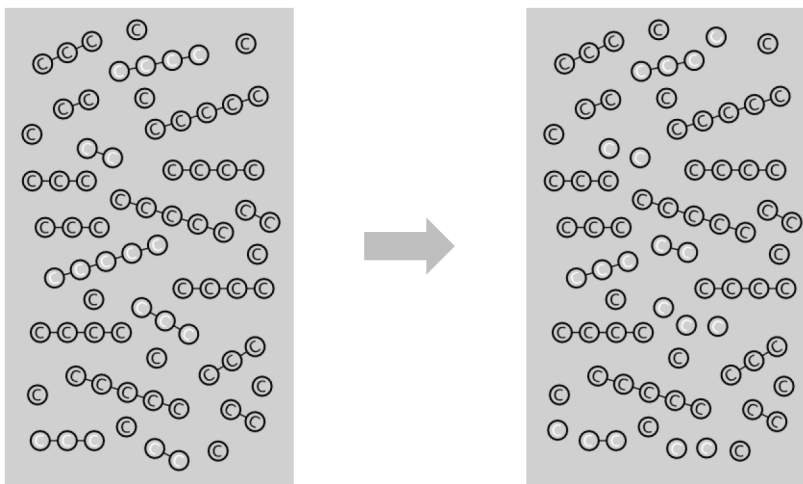


Figure 5—Final step towards a reservoir fluid composition at chemical reaction equilibrium, where the C₂-C₆ molecules are broken down into smaller molecules.

The systematics seen in the deviations between a measured and an equilibrium composition suggests that it might be possible to correct a simulated chemical equilibrium composition to have it agree with a measured non-equilibrium composition. If the mole percent of C₁ in the reservoir fluid were known, it would allow the deviation for C₁ to be distributed between the C₂-C₆ components. Based on data for 32 reservoir fluid compositions, it was found that an approximate mole percent of C₁ in a reservoir fluid can be simulated using the correlation in Equation (25) and the coefficients in Table 4.

$$C_1 \text{ mol\%} = B_1 + B_2 \times \rho + B_3 \times T \times \rho + \frac{B_4 \times P}{\rho} \quad (25)$$

The deviation between the molar amount of C₁ in the equilibrium composition and that simulated by Equation (25) is subsequently distributed between C₂ to C₆. Each of the C₃-C₆ mol% is corrected by a factor that is 1.5 times higher than the correction factor used for C₂. A calculation example is shown in Appendix B.

Table 4—Coefficients in Equation (25) for T in K, P in bar and ρ in g/cm³.

Coefficient	C ₁ mol%
B ₁	83.591
B ₂	-65.304
B ₃	-2.7263×10^{-2}
B ₄	1.2988×10^{-2}

Table 5—Reservoir fluid compositions from 4 neighboring formations in a North Sea field.

Component	Formation A			Formation B		Formation C		Formation D
	Fluid 1	Fluid 2	Fluid 3	Fluid 1	Fluid 2	Fluid 1	Fluid 2	Fluid 1
	Mol%	Mol%	Mol%	Mol%	Mol%	Mol%	Mol%	Mol%
N ₂	0.189	0.191	0.185	0.209	0.163	0.166	0.154	0.420
CO ₂	4.319	4.264	4.104	3.925	3.831	4.148	4.212	2.801
C ₁	74.403	74.835	71.166	73.710	73.508	74.603	74.956	55.150
C ₂	7.943	8.038	8.027	8.289	8.393	8.064	7.986	9.239
C ₃	3.511	3.464	3.930	3.679	3.775	3.527	3.439	6.571
iC ₄	0.573	0.582	0.607	0.605	0.615	0.586	0.574	1.139
nC ₄	1.171	1.102	1.373	1.202	1.249	1.140	1.107	2.760
iC ₅	0.465	0.455	0.532	0.484	0.498	0.467	0.454	1.048
nC ₅	0.537	0.501	0.651	0.545	0.592	0.522	0.508	1.390
C ₆	0.694	0.662	0.844	0.714	0.741	0.688	0.668	1.691
C ₇₊	6.195	5.909	8.577	6.644	6.639	6.091	5.94	17.795
C ₇	1.079	1.031	1.303	1.111	1.140	1.048	1.031	2.751
C ₈	1.165	1.121	1.408	1.216	1.226	1.145	1.114	2.926
C ₉	0.696	0.655	0.890	0.724	0.752	0.685	0.669	1.759
C ₁₀	0.495	0.457	0.656	0.513	0.515	0.482	0.472	1.341
C ₁₁	0.350	0.325	0.484	0.367	0.374	0.340	0.336	1.005
C ₁₂	0.281	0.262	0.394	0.296	0.294	0.270	0.269	0.843
C ₁₃	0.276	0.257	0.396	0.293	0.288	0.266	0.264	0.837
C ₁₄	0.232	0.218	0.337	0.249	0.245	0.227	0.224	0.718
C ₁₅	0.215	0.200	0.309	0.229	0.227	0.208	0.206	0.678
C ₁₆	0.167	0.158	0.245	0.178	0.178	0.162	0.159	0.530
C ₁₇	0.143	0.137	0.211	0.153	0.150	0.139	0.138	0.456
C ₁₈	0.132	0.127	0.198	0.142	0.141	0.127	0.127	0.449
C ₁₉	0.114	0.111	0.176	0.124	0.119	0.112	0.111	0.372
C ₂₀₊	0.850	0.850	1.570	1.049	0.990	0.880	0.820	3.130
C ₇₊ M	174.0	176.0	192.5	183.7	178.8	176.4	174.0	188.3
C ₇₊ density (g/cm ³)	0.8116	0.8111	0.8170	0.8043	0.8046	0.8015	0.7988	0.8175
Temp (°C)	149.1	150.0	151.1	152.6	153.0	145.9	147.7	156.8
Pressure (bar)	595.5	596.4	597.6	597.8	598.6	592.3	594.1	606.4
Depth (m)	3799.1	3823.2	3851.4	3888.7	3897.5	3721.3	3764.2	3993.2
H/C ratio	2.785	2.807	2.587	2.733	2.723	2.709	2.808	2.231
In-situ density (g/cm ³)	0.386	0.381	0.433	0.394	0.393	0.385	0.380	0.550

Analysis of North Sea Fluid Compositions

The outlined simulation tools have been tested on eight fluid compositions from four compartmentalized formations in a North Sea field. The compositions in three of the formations are similar, while the composition in the fourth formation is different. It is to be investigated whether the fluids in the three formations of similar compositions are at chemical reaction equilibrium at reservoir conditions. Table 4 shows the fluid compositions, the reservoir conditions, the depths at which the samples were taken, the H/C ratios and the in-situ reservoir fluid densities.

In Formation A three samples were taken over a depth interval of 52.3 meters. The concentration of components heavier than C_1 increases with depth. To clarify whether this is caused by compositional grading, the common equation of state model in Table C.1 in Appendix C was developed for the three fluid compositions for use with the volume-corrected SRK equation [12,13]. A depth gradient simulation was performed with Fluid 1 as the reference fluid and assuming a vertical temperature gradient of $0.04^\circ\text{C}/\text{m}$. Bar diagrams comparing measured and simulated fluid compositions sampled 24.1 m and 52.3 m below the reference sample are shown in Figure 6 and detailed in Table C.3 in Appendix C. The simulated compositions in these two depths agree nicely with those sampled. Formation A is therefore an example of a fluid system for which chemical reaction equilibrium will not be reached due to significant compositional grading.

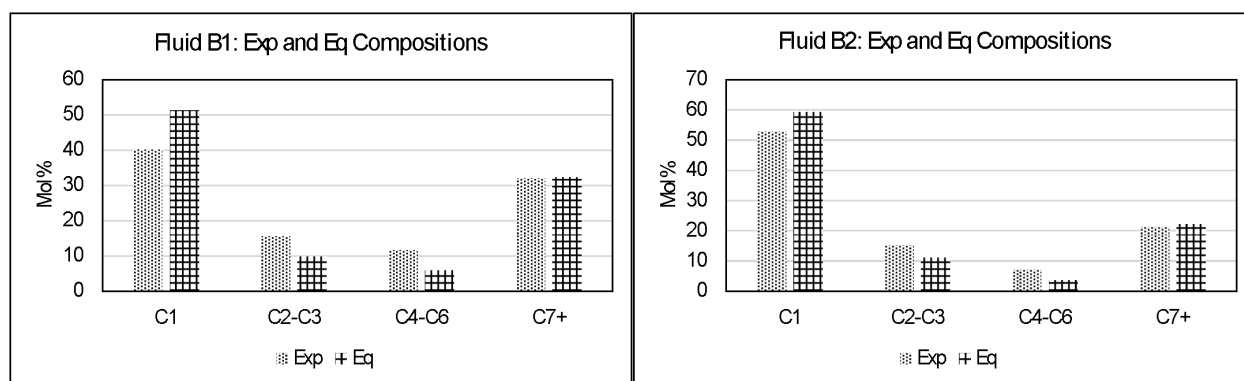


Figure 6—Bar diagrams showing measured and simulated compositions in depths of 3823.2 m and 3851.4 m in Formation A in Table 3 with the fluid composition in a depth of 3799.1 m as the reference fluid. The equation of state model is shown in Table C.1 in Appendix C. The simulation was carried out assuming a vertical temperature gradient of $0.04^\circ\text{C}/\text{m}$. Detailed compositions are shown in Table C.3 in Appendix C.

To quantify how much compositional grading affects the fluid compositions in Formation A, an equilibrium composition is calculated using average values for T , P , and H/C ratio for the three samples. A bar diagram is shown in Figure 7a comparing the simulated equilibrium composition with the 3 sampled compositions. The C_1 concentration in the simulated equilibrium composition is lower than in the two upper samples but higher than in the lower sample. Similarly, the simulated concentration of heavier components is higher than in the two upper samples and lower than in the lower sample. This means, the simulated chemical reaction equilibrium composition represents a kind of average composition for Formation A.

Two samples were taken in Formation B with a distance of 8.8 meters. As seen from Figure 7b, the two compositions are almost identical, which is also to be expected given the small distance. The diagram also shows that an almost identical fluid composition is simulated by assuming chemical reaction equilibrium.

In Formation C, two samples were taken with a distance of 42.9 meters. This is sufficient to expect a significant compositional difference if the two sampling points were in communication. However, as can be seen from Figure 7c, the compositions of the two samples are almost identical, which makes it unlikely that the two sampling sites should communicate. This is confirmed by the almost identical equilibrium composition simulated for the average sampling temperature, pressure, and H/C ratio.

While the fluids in the formations A, B, and C are at or close to chemical reaction equilibrium, this is not the case for the sample from Formation D. It is an oil sample taken at a somewhat higher depth than the other samples, which are all gas condensates. A comparison of the sampled composition with the one derived from chemical reaction equilibrium (Figure 7d) shows that an equilibrium composition would contain 62.46 mol% C_1 , while the sampled composition only contained 55.15 mol% C_1 . For all four formations, detailed simulated equilibrium compositions can be seen from Table C.4 in Appendix C.

The procedure described in the previous section for estimating the actual fluid composition by correcting the equilibrium composition was applied to the fluid in Formation D. As can be seen from [Table C.5](#) in [Appendix C](#), the estimated composition agrees well with the measured composition.

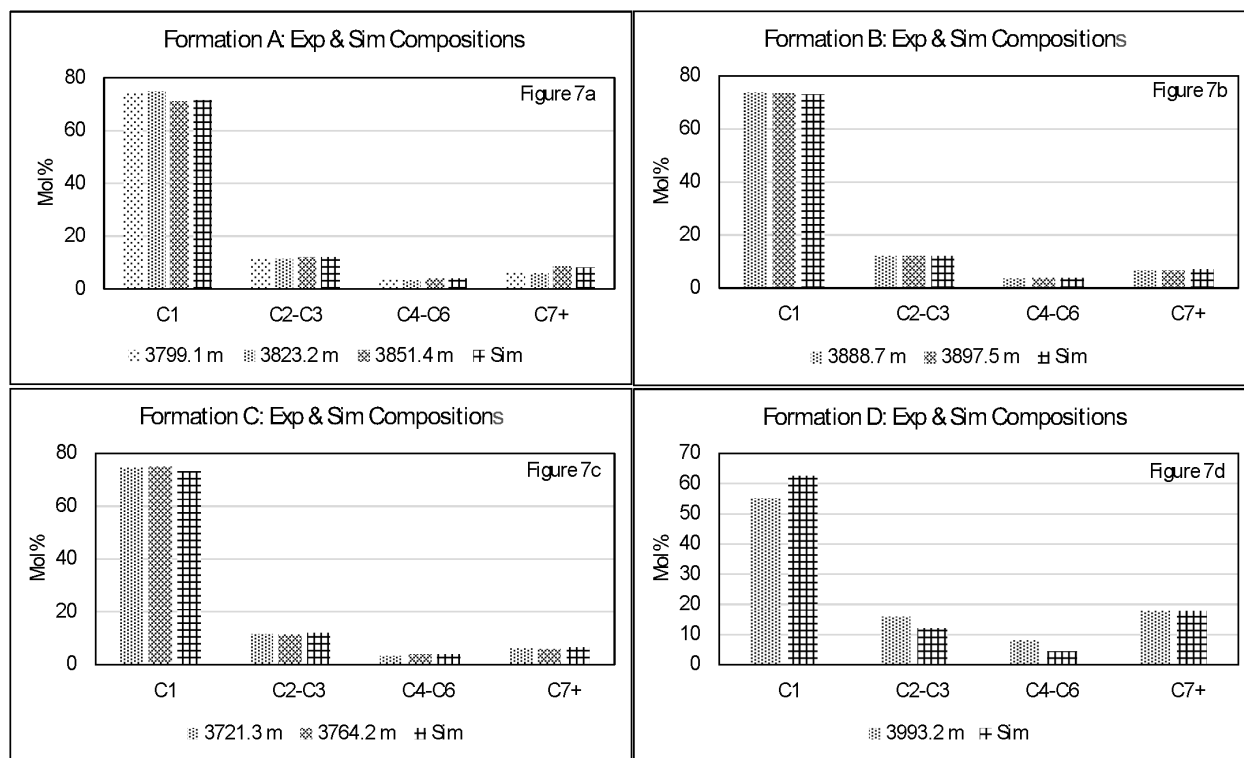


Figure 7—Bar diagrams comparing the measured fluid compositions in [Table 3](#) with the fluid compositions that would have existed in each of the four formations in the case of chemical reaction equilibrium. Detailed simulated compositions are shown in [Table C.4](#) in [Appendix C](#).

Conclusion

A procedure is presented for predicting the composition of a reservoir fluid that is in chemical reaction equilibrium in the reservoir. The method has reservoir pressure and temperature and the fluid density at reservoir conditions as the only required input. For some reservoir fluids, the equilibrium state cannot be reached due to gravitational forces and the vertical temperature gradient found in most reservoirs. For other fluids the decomposition process leading to the formation of oil and gas has not yet reached chemical reaction equilibrium. Such fluids will have a C_1 concentration that are lower than in the equilibrium composition, while the concentrations of C_2 – C_6 are correspondingly higher. If the reservoir conditions and the fluid density at these conditions are known, the simulated equilibrium composition can be corrected to provide a good estimate of the actual composition and thus provide an approximate reservoir fluid composition at the sampling stage, when a measured fluid composition is not yet available.

An analysis has been carried out of fluid data from a field in the North Sea. Almost identical fluid samples were taken from compartmentalized reservoir zones, which fluids were found to be in chemical reaction equilibrium. The material also included samples from a reservoir zone with a significant compositional variation with depth. For this zone, the simulated equilibrium fluid composition was found to be a kind of average fluid composition for the entire reservoir. A fluid sample taken further down contained an oil composition that had not yet reached chemical reaction equilibrium. For this fluid, it is shown how the actual fluid composition can be estimated by a simple correction of the equilibrium composition.

The presented simulation procedures have several applications. When it is unclear whether reservoir zones are separate or in communication, simulated chemical reaction equilibrium compositions, possibly

combined with simulations of the compositional variation with depth, may help to clarify this question. In the exploration phase, when a reliable composition is not yet available, an approximate composition can be estimated from reservoir pressure and temperature and the in-situ fluid density. It may also be possible to use equilibrium compositions in geochemical studies, where it may be of interest to know how far a fluid is from chemical reaction equilibrium.

Notation

A	Total number of atoms
A ₁ -A ₄	Coefficients in Equation (24)
a	Number of atoms
a ₁ -a ₄	Coefficients in expression for Gibbs energies of formation for C ₇₊ components
B ₁ -B ₄	Coefficients in Equation (25)
C	Carbon
C _n	Carbon number
C _p	Heat capacity
C ₁ -C ₄	Coefficients in 3 rd degree temperature polynomial for ideal gas heat capacity
Exp	Measured
Eq	Equilibrium
F	Function defined in Equation (4)
f	Fugacity
G	Gibbs free energy
g	Gravitational acceleration
H	Hydrogen or enthalpy
h	Depth
i	Component index
j	Component index
K	Total number of elements
k	Element number
M	Molecular weight
N	Number of species or number of components
n	Number of moles
P	Pressure
R	Gas constant
Sim	Simulated
SRK	Soave-Redlich-Kwong
T	Temperature
z	Mole fraction

Greek letters

φ	Fugacity coefficient
λ	Lagrange multiplier
μ	Chemical potential

Sub and super scripts

abs	Absolute
C	Carbon
f	Formation

H	Hydrogen
h	Depth
i	Component number
ig	Ideal gas
k	Atom index
ref	Reference
res	Residual
t	Total
◦	Reference
~	Partial

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Appendix A

Table A.1—Reservoir fluid compositions assessed to be at chemical reaction equilibrium at reservoir conditions.

Component	Fluid A1 [14]			Fluid A2 [14]			Fluid A3 [15]			Fluid A4		
	Exp Mol%	Eq Mol%	%Dev	Exp Mol%	Eq Mol%	%Dev	Exp Mol%	Eq Mol%	%Dev	Exp Mol%	Eq Mol%	%Dev
N ₂	0.848	(0.848)	-	0.530	0.530	0.0	0.340	(0.340)	-	0.530	(0.530)	-
CO ₂	0.649	(0.649)	-	3.300	3.300	0.0	3.590	(3.590)	-	1.460	(1.460)	-
C ₁	83.583	83.167	-0.5	72.970	72.923	-0.1	67.421	69.048	2.4	56.300	55.935	-0.6
C ₂	5.948	6.223	4.6	7.680	8.104	5.5	9.020	8.297	-8.0	6.980	6.582	-5.7
C ₃	2.911	3.046	4.7	4.100	3.811	-7.0	4.310	4.195	-2.7	3.840	3.629	-5.5
iC ₄	0.448	0.462	3.2	0.700	0.659	-5.9	0.930	0.757	-18.6	0.620	0.645	4.0
nC ₄	1.108	1.123	1.4	1.420	1.329	-6.4	1.710	1.574	-7.9	1.630	1.490	-8.6
iC ₅	0.361	0.362	0.3	0.540	0.532	-1.5	0.740	0.658	-11.1	0.650	0.593	-8.7
nC ₅	0.481	0.494	2.6	0.670	0.626	-6.6	0.850	0.787	-7.5	0.850	0.770	-9.4
C ₆	0.599	0.547	-8.7	0.850	0.839	-1.3	1.380	1.099	-20.3	1.530	1.063	-30.5
C ₇₊	3.064	3.078	0.5	7.24	7.348	1.5	9.709	9.655	-0.6	25.61	27.304	6.6
C ₇	0.801	0.952	18.8	1.330	1.152	-13.4	1.500	1.244	-17.0	1.860	4.428	138.1
C ₈	0.761	0.644	-15.4	1.330	0.971	-27.0	1.690	1.090	-35.5	1.550	3.606	132.6
C ₉	0.471	0.47	-0.2	0.780	0.855	9.6	1.140	0.990	-13.2	2.050	3.174	54.8
C ₁₀	0.307	0.325	5.8	0.610	0.724	18.6	0.800	0.866	8.2	2.110	2.643	25.3
C ₁₁	0.216	0.22	2.0	0.420	0.604	43.9	0.635	0.747	17.6	1.290	2.166	67.9
C ₁₂	0.151	0.151	0.1	0.330	0.508	53.8	0.547	0.647	18.4	1.420	1.807	27.3
C ₁₃	0.106	0.102	-3.7	0.420	0.422	0.5	0.471	0.556	18.1	1.330	1.495	12.4
C ₁₄	0.075	0.07	-6.9	0.240	0.354	47.6	0.406	0.482	18.8	1.340	1.264	-5.7
C ₁₅	0.052	0.048	-7.0	0.300	0.300	0.0	0.349	0.422	20.8	1.190	1.094	-8.1
C ₁₆	0.037	0.033	-10.7	0.170	0.253	48.6	0.301	0.368	22.2	1.030	0.945	-8.3
C ₁₇	0.026	0.022	-16.1	0.210	0.209	-0.7	0.259	0.316	22.1	1.110	0.793	-28.6
C ₁₈	0.018	0.014	-22.6	0.150	0.169	12.8	0.223	0.268	20.1	0.870	0.647	-25.7
C ₁₉	0.013	0.008	-35.1	0.150	0.133	-11.4	0.192	0.221	15.1	0.700	0.499	-28.7
C ₂₀₊	0.030	0.019	-36.2	0.800	0.694	-13.2	1.196	1.438	20.3	7.760	2.743	-64.7
Temp (°C)	96.4			143.0			139.9			122.0		
Pressure (bar)	350			575			650			369		
H/C ratio	3.215			2.712			2.535			2.074		
In-situ density g/cm ³	0.259			0.399			0.464			0.640		

Appendix B

Table B.1—Reservoir fluid compositions not at chemical reaction equilibrium at reservoir conditions.

Component	Fluid B1 [15]				Fluid B2 [15]			
	Exp Mol%	Eq Mol%	%Dev	Estimated Non-Eq Mol% ^{*1)}	Exp Mol%	Eq Mol%	%Dev	Estimated Non-Eq Mol% ^{*1)}
N ₂	0.390	(0.390)	-	(0.390)	0.690	(0.69)	-	(0.69)
CO ₂	0.300	(0.300)	-	(0.300)	3.138	(3.138)	-	(3.138)
C ₁	40.200	51.261	27.5	39.560	52.792	59.299	12.3	50.359
C ₂	7.610	6.004	-21.1	7.985	8.866	7.620	-14.1	9.852
C ₃	7.950	3.890	-51.1	7.761	6.296	3.489	-44.6	6.767
iC ₄	1.190	0.706	-40.7	1.408	1.060	0.621	-41.4	1.204
nC ₄	4.080	1.891	-53.7	3.773	2.480	1.177	-52.6	2.283
iC ₅	1.390	0.742	-46.7	1.480	0.870	0.499	-42.7	0.968
nC ₅	2.150	1.068	-50.3	2.131	1.170	0.559	-52.2	1.084
C ₆	2.790	1.472	-47.2	2.937	1.450	0.795	-45.2	1.542
C ₇₊	31.950	32.227	1.0	32.227	21.191	22.112	4.3	22.112
C ₇	4.280	5.352	25.0	5.352	2.292	2.74	19.5	2.74
C ₈	4.310	4.309	0.0	4.309	2.612	2.381	-8.9	2.381
C ₉	3.080	3.775	22.6	3.775	1.802	2.171	20.5	2.171
C ₁₀	2.470	3.116	26.1	3.116	1.770	1.901	7.4	1.901
C ₁₁	1.910	2.530	32.5	2.530	1.198	1.644	37.2	1.644
C ₁₂	1.690	2.099	24.2	2.099	1.160	1.434	23.7	1.434
C ₁₃	1.590	1.728	8.7	1.728	0.993	1.242	25.1	1.242
C ₁₄	1.220	1.461	19.8	1.461	1.040	1.089	4.8	1.089
C ₁₅	1.250	1.271	1.7	1.271	0.881	0.969	10.0	0.969
C ₁₆	1.000	1.106	10.6	1.106	0.713	0.86	20.6	0.86
C ₁₇	0.990	0.933	-5.7	0.933	0.619	0.749	21.0	0.749
C ₁₈	0.920	0.765	-16.9	0.765	0.701	0.641	-8.6	0.641
C ₁₉	0.600	0.589	-1.9	0.589	0.592	0.53	-10.6	0.53
C ₂₀₊	6.640	3.243	-51.2	3.243	4.818	3.761	-21.9	3.761
Temp (°C)	97.5				164			
Pressure (bar)	393				486			
H/C ratio	2.010				2.416			
In-situ density (g/cm ³)	0.683				0.573			

^{*1)} Details below

Estimation of actual B1 composition:

$$\text{Equation (25): } C_1 \text{ mol\%} = 83.59 - 65.304 \times 0.683 - 2.7263 \times 10^{-2} \times 370.65 \times 0.683 + \frac{1.2988 \times 10^{-2} \times 393}{0.683} = 39.56$$

$$\text{Equilibrium } C_1 \text{ mol\%} - \text{Estimated } C_1 \text{ mol\%} = 51.26 - 39.56 = 11.70$$

$$\text{Correction factor for } C_2 \text{ mol\%} = \frac{6.004 + 3.890 \times 1.5 + 3.890 \times 1.5 + 0.706 \times 1.5 + 1.891 \times 1.5 + 0.742 \times 1.5 + 1.068 \times 1.5 + 1.472 \times 1.5}{6.004 + 3.890 + 3.890 + 0.706 + 1.891 + 0.742 + 1.068 + 1.472 + 11.7} = 1.330$$

$$\text{Correction factor for } C_3 - C_6 \text{ mol\%'s} = 1.330 \times 1.5 = 1.995$$

Appendix C

Table C.1—Common equation of state model for the three fluid samples from Formation A in Table 3. The model is developed for use with the volume corrected SRK equation [12,13].

Component	Critical Temp °C	Critical Pressure bar	Acentric Factor	Volume Correction cm ³ /mol	H ^{ig} at 273.15 K J/g
N ₂	-146.95	33.94	0.040	0.92	294.6
CO ₂	31.05	73.76	0.225	3.03	438.0
C ₁	-82.55	46.00	0.008	0.63	0.0
C ₂	32.25	48.84	0.098	2.63	321.5
C ₃	96.65	42.46	0.152	5.06	438.5
iC ₄	134.95	36.48	0.176	7.29	499.0
nC ₄	152.05	38.00	0.193	7.86	499.0
iC ₅	187.25	33.84	0.227	10.93	535.9
nC ₅	196.45	33.74	0.251	12.18	535.9
C ₆	234.25	29.69	0.296	17.98	560.9
C ₇	267.42	30.43	0.468	14.15	551.4
C ₈	286.45	27.58	0.500	21.07	562.7
C ₉	308.01	24.62	0.540	30.02	574.2
C ₁₀	326.41	22.62	0.576	37.19	582.7
C ₁₁	343.50	21.07	0.612	43.33	589.7
C ₁₂	360.64	19.73	0.650	49.02	596.0
C ₁₃	376.76	18.66	0.688	53.52	601.3
C ₁₄	393.06	17.72	0.727	57.33	606.1
C ₁₅	409.50	16.89	0.768	60.29	610.4
C ₁₆	425.19	16.23	0.807	61.97	614.1
C ₁₇	439.37	15.73	0.844	62.28	617.2
C ₁₈	452.19	15.36	0.877	61.53	619.7
C ₁₉	462.97	15.12	0.904	59.84	621.6
C ₂₀₊	637.71	13.56	1.167	15.60	637.9

Table C.2—Non-zero binary interactions parameters for use with the equation of state model in Table C.1.

	N ₂	CO ₂
CO ₂	-0.0315	
C ₁	0.0278	0.1200
C ₂	0.0407	0.1200
C ₃	0.0763	0.1200
iC ₄	0.0944	0.1200
nC ₄	0.0700	0.1200
iC ₅	0.0867	0.1200
nC ₅	0.0878	0.1200
C ₆	0.0800	0.1200
C ₇₊	0.0800	0.1000

Table C.3—Simulated mol% compositions in depths of 3823.2 m and 3851.4 m in the North Sea Formation A with the fluid composition sampled in a depth of 3799.1 m as reference fluid. The volume corrected SRK-equation is used with the model parameters in [Tables C.1](#) and [C.2](#). The vertical temperature gradient is assumed to be 0.04°C/m. Measured compositions can be seen from [Table 3](#).

Component	3823.2 m	3851.4 m
N ₂	0.185	0.173
CO ₂	4.335	4.315
C ₁	73.650	71.233
C ₂	8.000	8.051
C ₃	3.582	3.703
iC ₄	0.590	0.622
nC ₄	1.211	1.286
iC ₅	0.486	0.525
nC ₅	0.562	0.611
C ₆	0.734	0.815
C ₇₊	6.665	8.665
C ₇	1.153	1.319
C ₈	1.254	1.453
C ₉	0.756	0.889
C ₁₀	0.542	0.647
C ₁₁	0.386	0.468
C ₁₂	0.313	0.385
C ₁₃	0.310	0.388
C ₁₄	0.263	0.336
C ₁₅	0.247	0.321
C ₁₆	0.194	0.257
C ₁₇	0.168	0.228
C ₁₈	0.156	0.217
C ₁₉	0.136	0.192
C ₂₀₊	0.787	1.565

Table C.4—Mol% compositions for North Sea fluid in Table 3 assuming chemical reaction equilibrium at reservoir conditions. For Formations A, B, and C the equilibrium compositions are simulated using average reservoir temperature, pressure, and H/C ratio of the samples taken.

Component	Formation A	Formation B	Formation C	Formation D
N ₂	(0.188)	(0.186)	(0.160)	(0.420)
CO ₂	(4.229)	(3.878)	(4.180)	(2.801)
C ₁	71.611	72.980	73.163	62.458
C ₂	8.251	8.321	8.172	8.177
C ₃	3.805	3.766	3.774	3.957
iC ₄	0.664	0.652	0.652	0.723
nC ₄	1.298	1.260	1.292	1.414
iC ₅	0.531	0.512	0.520	0.612
nC ₅	0.610	0.584	0.603	0.699
C ₆	0.829	0.791	0.809	1.009
C ₇₊	7.985	7.07	6.676	17.731
C ₇	1.168	1.080	1.055	1.900
C ₈	1.000	0.919	0.891	1.701
C ₉	0.890	0.811	0.782	1.578
C ₁₀	0.764	0.691	0.661	1.411
C ₁₁	0.646	0.580	0.552	1.247
C ₁₂	0.550	0.489	0.462	1.108
C ₁₃	0.462	0.409	0.384	0.977
C ₁₄	0.392	0.344	0.321	0.871
C ₁₅	0.336	0.291	0.270	0.784
C ₁₆	0.286	0.245	0.226	0.706
C ₁₇	0.240	0.204	0.187	0.626
C ₁₈	0.198	0.167	0.152	0.548
C ₁₉	0.159	0.133	0.120	0.467
C ₂₀₊	0.894	0.707	0.613	3.807

Table C.5—Measured (Exp), chemical equilibrium (Eq) composition and estimated non-equilibrium composition for the fluid in Formation D. The estimated I derived from reservoir pressure, temperature, and in-situ density.

Component	Exp Mol%	Eq Composition Mol%	Estimated Non-Eq Mol% ⁽¹⁾	%Dev Estimated & Exp Compositions
N ₂	0.420	(0.420)	(0.420)	-
CO ₂	2.801	(2.801)	(2.801)	-
C ₁	55.150	62.458	55.554	0.7
C ₂	9.239	8.177	9.237	0.0
C ₃	6.571	3.957	6.705	2.0
iC ₄	1.139	0.723	1.225	7.6
nC ₄	2.760	1.414	2.396	-13.2
iC ₅	1.048	0.612	1.037	-1.0
nC ₅	1.390	0.699	1.184	-14.8
C ₆	1.691	1.009	1.710	1.1
C ₇₊	17.795	17.731	17.731	-0.4
C ₇	2.751	1.900	1.900	-30.9
C ₈	2.926	1.701	1.701	-41.9
C ₉	1.759	1.578	1.578	-10.3
C ₁₀	1.341	1.411	1.411	5.2
C ₁₁	1.005	1.247	1.247	24.1
C ₁₂	0.843	1.108	1.108	31.4
C ₁₃	0.837	0.977	0.977	16.7
C ₁₄	0.718	0.871	0.871	21.3
C ₁₅	0.678	0.784	0.784	15.6
C ₁₆	0.530	0.706	0.706	33.2
C ₁₇	0.456	0.626	0.626	37.3
C ₁₈	0.449	0.548	0.548	22.0
C ₁₉	0.372	0.467	0.467	25.5
C ₂₀₊	3.130	3.807	3.807	21.6

⁽¹⁾ Calculation example in Appendix B